MASOUD SHIRZAD

Postdoctoral Researcher · Mechanical & Robotics Engineering

Research Profile

Researcher specializing in bioinspired and mechanical metamaterials, additive manufacturing, computational mechanics, and AI-assisted structural design for robotic and biomedical applications.

Design & Fabrication: Bioinspired metamaterials, 3D printing, biofabrication

Computational: Finite element analysis, machine learning, structural optimization

Experimental: Mechanical characterization, fatigue testing, experimental mechanics

Research Interests: Multifunctional metamaterials, intelligent structures, AI-assisted design

EDUCATION

Ph.D.

Mar 2022 — Feb 2026

Pukyong National University

Industry 4.0 Convergence Bionics Engineering — Meta-structures: a novel approach to biomimicry of natural tissues for biomedical applications.

M.Sc.

Sep 2014 — Dec 2016

Babol Noshirvani University of Technology

Mechanical Engineering — Optimal Design and Fabrication of Scaffold with 3D Printing Method.

B.Sc.

Sep 2010 — Sep 2014

Babol Noshirvani University of Technology

Mechanical Engineering — Investigation of 3D Scaffolds with Finite Element Method, Comparison to Experimental Data.

Research Experience

InnoCORE Fellow

Aug 2026 — Present

DGIST

Postdoctoral Researcher

Mar 2026 — Jul 2026

Pukyong National University

INTERESTS

- Metamaterials designs
- Machine Learning for structural designs
- Finite Element Methods
- Additive Manufacturing
- Experimental Mechanics

LANGUAGES

Persian

Native

English

Fluent · TOEFL iBT 99

Korean

Basic

SKILLS

Programming

Python · MATLAB · Design-Expert

CAD & Design

SOLIDWORKS

FEA

ABAQUS · COMSOL

Additive Manufacturing

FDM · DLP · SLA

PUBLICATIONS

1. **Masoud Shirzad**, J. Kang, M. Bodaghi, S.Y. Nam. "Bioinspired Hierarchical-Gradient Designs for Tendon-to-Bone Interfaces: Continuous and Discontinuous Fatigue Behavior."

   Materials & Design. · *2026*

2. **Masoud Shirzad**, M. Kordi, D. Kim, J. Choi, P. Ranganathan, J. Han, H.G. Lim, S.Y. Nam. "Anisotropic Tissue Biofabrication: Conventional and Advanced Fabrication Strategies for Functional Scaffolds."

   Biomedical Materials. · *2026*

3. M. Mohammadhosseinzadeh, **Masoud Shirzad**, P. Kahhal, D.H. Kwak, H. Ghorbani-Menghari. "Mechanical evaluation of bio-inspired meta-structures: insights from experimental and computational analysis."

   Int. J. Adv. Manuf. Technol. · *2026*

4. A. Zolfagharian, et al. "Roadmap on Artificial Intelligence-Augmented Additive Manufacturing."

   Advanced Intelligent Systems · *2026*

5. P. Ranganathan, et al. "Integrated Biofabrication of Artificial Esophageal Scaffolds using Electrospinning, Embedded DLP, and Extrusion Techniques."

   Materials Today Bio · *2025*

6. **Masoud Shirzad**, J. Kang, M. Bodaghi, S.Y. Nam. "Design and Prediction of Soft-to-Hard Transitions Using Bioinspired Hierarchical-Gradient Designs and Hybrid Stacking Machine Learning."

   Computers in Biology and Medicine · *2025*

7. **Masoud Shirzad**, A. Zolfagharian, S.Y. Nam, M. Bodaghi. "Comprehensive Review on Hybrid Lattice Meta-Structures for Biomedical Engineering Applications."

   European Journal of Mechanics - A/Solids · *2025*

8. J. Kang, **Masoud Shirzad**, P. Ranaganathan, D. Oh, S. Mundal, J. Oh, S.Y. Nam. "Suspended 3D Printing of Polycaprolactone/Hydroxyapatite composites for mimicking complex Bone Scaffolds."

   International Journal of Bioprinting · *2025* · *Co-first author*

9. **Masoud Shirzad**, M. Jafari Chashmi, S. Khakzadkelarijani, M. Bodaghi, S.Y. Nam. "Machine Learning-Assisted Design and Optimization of Auxetic Structures: A Bioinspired Approach to Mimic Natural Tissues."

   Advanced Engineering Materials · *2025*

10. J. Kang, **Masoud Shirzad**, J.M. Seok, W.-K. Jung, S.Y. Nam. "Mechanical enhancement of porous scaffolds through integration of auxetic and conventional structures."

    Materials Letters · *2024*

11. **Masoud Shirzad**, J. Kang, G. Kim, M. Bodaghi, S.Y. Nam. "Bioinspired 3D-Printed auxetic structures with enhanced fatigue behavior."

    Advanced Engineering Materials · *2024*

PUBLICATIONS (CONTINUED)

12 **Masoud Shirzad**, M. Bodaghi, D. Oh, M. Yi, S.Y. Nam. "Design and Optimization of Bioinspired Auxetic Structure for Biomedical Applications."

European Journal of Mechanics - A/Solids · 2024

13 D. Oh, **Masoud Shirzad**, M.C. Kim, E.-J. Chung, S.Y. Nam. "Rheology-informed hierarchical machine learning model for the prediction of printing resolution in extrusion-based bioprinting."

International Journal of Bioprinting · 2023

14 **Masoud Shirzad**, A. Zolfagharian, M. Bodaghi, S.Y. Nam. "Auxetic metamaterials for bone-implanted medical devices: recent advances and new perspectives."

European Journal of Mechanics - A/Solids · 2022

15 **Masoud Shirzad**, A. Zolfagharian, A. Matbouei, M. Bodaghi. "Design, evaluation, and optimization of 3D printed truss scaffolds for bone tissue engineering."

J. Mech. Behav. Biomed. Mater. · 2021

16 M. Jafari Chashmi, A. Fathi, **Masoud Shirzad**, R.A. Jafari-Talookolaei, M. Bodaghi, S.M. Rabiee. "Design and analysis of porous functionally graded femoral prostheses with improved stress shielding."

Designs · 2020

17 **Masoud Shirzad**, A. Matbouei, A. Fathi, S.M. Rabiee. "Experimental and numerical investigation of polymethyl methacrylate scaffolds for bone tissue engineering."

Proc. IMechE Part L · 2020

18 **Masoud Shirzad**, A. Fathi, S.M. Rabiee, S. Ghaffari, E. Zabihi-Neishabouri. "Three-dimensional printing of truss-like structure for use in scaffold: experimental, numerical, and analytical analyses."

Proc. IMechE Part C · 2020

19 A. Matbouei, A. Fathi, S.M. Rabiee, **Masoud Shirzad**. "Layered manufacturing of a three-dimensional polymethyl methacrylate (PMMA) scaffold used for bone regeneration."

Materials Technology · 2019

20 **Masoud Shirzad**, S.M. Rabiee. "Fabrication and Analysis of a 3D truss Pattern Scaffold made by 3D printing Method."

Iranian J. Mech. Eng. (ISME) · 2018

CONFERENCES & PRESENTATIONS

Poster	K-TERMS 2025 · Best Poster Award	*2025*
Oral	International Conference on Biofabrication 2024	*2024*
Poster	World Biomaterials Conference (WBC) 2024	*2024*
Poster	4D Materials Design and Additive Manufacturing Conf.	*2022*
Oral	TERMIS-AP 2022	*2022*

HONORS & AWARDS

- **Brain Korea Scholarship** — *2022*
- **Fire Grant** — *2023, 2025*
- **Best Poster Award · K-TERMS** — *2025*
- **RA / TA Scholarship** — *2022*
- **Blue Scholarship** — *2023, 2026*

RESEARCH & TEACHING

Research Assistant
Mar 2022 — Feb 2026

Pukyong National University · Integrated Biofabrication Lab

Teaching Assistant
Sep 2014 — Jun 2015

Babol Noshirvani University of Technology